

Marketing Campaign Performance

1. Project Overview

This project builds an interactive advertising performance dashboard in Microsoft Excel, covering the full analyst workflow: data cleaning, modeling, DAX measures, and dashboard design. It is part of an ongoing data analyst portfolio.

Tool stack	Excel: Power Query, Power Pivot (DAX), PivotTables, PivotCharts
Dataset	Ad performance data: Google Ads and Facebook Ads (spend, impressions, clicks, link clicks)
Date range	October 16, 2019 - July 7, 2020
Raw size	16,834 rows / 14 columns

2. Data Cleaning (Power Query)

Data quality issues identified during profiling and resolved in Power Query before loading to the data model:

- Removed 205 fully duplicate rows.
- Standardized placeholder values ('-') in audience_type, creative_type, and creative_name to proper N/A.
- Corrected ~1,733 rows where the device column literally contained the text "device" instead of a real value (Desktop / Mobile / Tablet), relabeled as Unknown
- Dropped the product and phase columns, which held a single constant value across all rows and added no analytical value
- Enforced correct data types: Date as Date, spend/link_clicks as Decimal Number, impressions/clicks as Whole Number

3. DAX Measures (Power Pivot)

The following measures were built in the Data Model to drive all PivotTables and Pivot Charts:

Measure	Formula	Result
Total Spend	<code>SUM(spend)</code>	2,503,115.19\$
Total Impressions	<code>SUM(impressions)</code>	4,847,190
Total Clicks	<code>SUM(clicks)</code>	201,633
CTR	<code>DIVIDE(SUM(clicks), SUM(impressions))</code>	4.16%
CPC	<code>DIVIDE(SUM(spend), SUM(clicks))</code>	12.41\$
CPM	<code>DIVIDE(SUM(spend), SUM(impressions)) * 1000</code>	516.41\$

4. Dashboard Design

A single Dashboard sheet consolidates four Pivot Charts built from the underlying Pivot Tables:

- Trend view - Line chart of Total Spend and Total Clicks over time
- Platform comparison - Clustered column chart of Total Spends , Impressions and Clicks by Platform and Device type.
- Demographic breakdown - Stacked Column chart of Total Spend by age group and device type.

5. Skills Demonstrated

- Data cleaning and transformation with Power Query (M)
- Data modeling and calculated measures with Power Pivot (DAX)
- Pivot Table / Pivot Chart design for multi-dimensional analysis
- Dashboard layout, and Report for a portfolio audience

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